Xingyu Chen

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Education

Boston University — Boston, MA
Ph.D. in Computer Engineering (in progress) — *Sep. 2024–Present*
Research focus: distributed networks, blockchain systems, and data-efficient protocols

Boston University — Boston, MA
Master of Engineering in Electrical and Computer Engineering — *Sep. 2022–Dec. 2023*

Fudan University — Shanghai, China
Bachelor of Science in Microelectronics — *Sep. 2018–Jun. 2022*

Publications

1. **Out-of-Band Transaction Pool Sync for Large Dynamic Blockchain Networks**
   Novak Boskov, **Xingyu Chen**, Sevval Simsek, Ari Trachtenberg, and David Starobinski.
   International Journal of Network Management, Mar. 2024.
   doi:10.1002/nem.2265

2. **Scaling the Lightning Network with Practical Set Reconciliation**
   Xingyu Chen, Anish Sinha, Ari Trachtenberg, and David Starobinski.
   IEEE International Conference on Blockchain and Cryptocurrency (ICBC), Jun. 2026.
   doi:10.1109/ICBC67748.2026.11575524

3. **Network-Aware Modeling and Evaluation of Set Reconciliation Protocols**
   Xingyu Chen, Ari Trachtenberg, and David Starobinski.
   ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS), under review, 2026.

4. **Isolation Failure from Shared Storage: Characterizing and Exploiting Page-Cache SCA Leakage Across Containers and VMs**
   Alon Abudraham, **Xingyu Chen**, Itamar Levi, and Ari Trachtenberg.
   IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES), under review, 2026.
   doi:10.48550/arXiv.2607.17518

Research Projects

Lightning Network Scaling with Practical Set Reconciliation

Bitcoin · Lightning Network · Set Reconciliation · C++ · TCP · Network Measurement

- Investigated the scalability of the **Bitcoin Lightning Network (LN)** gossip protocol through **packet-level measurements**, identifying substantial redundant communication caused by flooding-based synchronization.
- Reformulated Lightning gossip synchronization as a **set-reconciliation** problem, allowing peers with highly overlapping network states to exchange only their differences.
- Integrated the **GenSync C++** reconciliation framework with **Core Lightning** to evaluate **CPI-Sync, Interactive CPI, Bloom filters, IBLT, MET-IBLT, and Rateless IBLT** under realistic network conditions.
- Designed and implemented **Adaptive-IBLT**, which progressively adjusts IBLT size when the set difference is unknown and reuses decoded elements through **partial decoding**.
- Evaluated the protocols using real **Lightning gossip snapshots**, demonstrating that reconciliation can reduce synchronization time from hours to minutes while substantially reducing redundant traffic.

Network-Aware Modeling and Evaluation of Set Reconciliation Protocols

Distributed Systems · TCP/IP · Network Modeling · C++ · Mininet · Wireshark

- Developed a **transport-aware analytical framework** for predicting practical set-reconciliation performance from communication volume, interaction depth, latency, bandwidth, and **TCP flow-control** behavior.
- Extended **GenSync** to benchmark reconciliation protocols under configurable **RTT, bandwidth, packet loss, and set-difference** conditions, enabling reproducible evaluation beyond asymptotic communication complexity.
- Characterized **latency-, bandwidth-, and computation-bound regimes**, showing that protocols with similar communication complexity can exhibit substantially different end-to-end performance.

- Systematically evaluated **CPI-Sync, Interactive CPI, IBLT, MET-IBLT, Rateless IBLT, and Adaptive-IBLT** across network and workload conditions.
- Proved a **constant-factor communication-cost bound** for Adaptive-IBLT relative to an ideal IBLT with perfect prior knowledge of the set difference.
- Replayed reconciliation workloads using deployment-derived **Lightning Network** parameters, demonstrating an approximately **83% reduction in combined transport overhead** relative to native flooding-based synchronization.

Page-Cache Side-Channel Leakage Across Containers and Virtual Machines
Systems Security · Side Channels · Linux · Docker · gVisor · Kata Containers · QEMU/KVM

- Investigated the **Linux page cache** as an **OS-mediated timing side channel**, demonstrating that shared storage state can leak information across container, sandbox, and virtual-machine isolation boundaries.
- Developed **unprivileged timing techniques** for inferring page-cache residency without explicit kernel interfaces such as `mincore()`.
- Evaluated leakage across **Docker, gVisor (systrap and KVM), Kata Containers, Firecracker, Cloud Hypervisor, and QEMU/KVM** under multiple filesystem, virtualization, and storage configurations.
- Characterized how **OverlayFS, virtio-fs, QCOW2 images, loop-backed devices, direct I/O, and dedicated block devices** affect cross-isolation leakage and signal-to-noise ratio.
- Demonstrated that **virtualization** can attenuate and reshape timing signals but does not inherently eliminate leakage when workloads access shared host-cacheable storage objects.
- Built a cross-isolation **covert channel** to quantify page-cache leakage capacity and reliability under **Docker and gVisor**.
- Developed a **WordPress/MySQL side-channel case study** showing that page-cache access patterns can reveal coarse-grained application behavior, including login, logout, password reset, post publication, user management, and homepage access.

Ongoing Research

Prompt Leakage from Mixture-of-Experts LLM Page-Cache Traces
LLM Security · Mixture-of-Experts · Page-Cache Side Channels · Qwen · llama.cpp · GGUF

- Investigating whether input-dependent **Mixture-of-Experts (MoE) routing** leaks semantic information through the **Linux page-cache access patterns** of shared model-weight files.
- Built an automated measurement pipeline for **Qwen3-30B-A3B** running with **llama.cpp/llama-server**, combining controlled cache eviction, memory pressure, `mincore()` snapshots, and block-level **GGUF** residency analysis.
- Demonstrated prompt-dependent cache fingerprints across unseen prompts: traces exhibit substantially higher similarity within the same prompt or semantic class than across classes, and page-level spatial features support semantic prompt classification.
- Developing an end-to-end attack pipeline that maps **GGUF file offsets to MoE layers and experts**, recovers noisy expert-routing traces from page-cache observations, and uses the recovered routing information to reconstruct or characterize victim prompts.
- Evaluating realistic shared-model deployments across **llama.cpp, Ollama, and vLLM**, with emphasis on mutually untrusted inference workloads that share immutable host-backed model weights.

Technical Skills

Programming Languages	C++, Python, Java, Go, JavaScript, SQL, MATLAB, HTML/CSS
Systems & Security	Linux, Docker, gVisor, Kata Containers, QEMU/KVM, Firecracker, Cloud Hypervisor, page cache, side-channel analysis, virtualization, OverlayFS, virtio-fs
Networking & Distributed Systems	TCP/IP, set reconciliation, distributed systems, Bitcoin, Lightning Network, network measurement, packet analysis, Mininet
LLM Models & Inference	Llama, Qwen, Gemma, Mixture-of-Experts (MoE), llama.cpp, GGUF, model quantization
Data Structures & Protocols	Invertible Bloom Lookup Tables (IBLT), Bloom filters, Characteristic Polynomial Interpolation (CPI), rateless coding
Web & Backend	Spring, Spring MVC, Hibernate, React, REST APIs, JWT, MySQL, Elasticsearch
Cloud & Development Tools	Google Cloud Platform, AWS, Git/GitHub, Wireshark, LaTeX